Nature-Style Technical Report

Quantum Phase-Resolved Dual-MRF Pulse Sequences for Cardiac and Stroke Imaging with Interoperative MR-Guided Tumour Resection Robotics

Authors: NeuroMorph Imaging Systems Group

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Abstract

We present an integrated pulse-sequence framework combining cardiac cine imaging, acute stroke tissue characterization, and interoperative MR-guided tumour resection. The framework introduces elliptic modulo resonators for quantum phase-resolved RF control and dual MR fingerprinting (Dual-MRF) to jointly estimate relaxation and transport biomarkers. We derive finite reconstruction operators extending beyond classical partial Fourier methods by introducing fractional boundary gaps in k-space closure constraints. Simulation-grade sequence specification is released in a scanner-style .seq file for translational prototyping.

1. Clinical-Technical Motivation

A unified protocol is useful when a centre performs:

1. Cardiac viability and flow mapping.

2. Hyperacute stroke core-penumbra triage.

3. Intraoperative tumour resection with robotic guidance under MR thermometry.

Conventional workflows split these tasks across distinct pulse families, increasing latency and calibration overhead. We instead use one quantum phase-resolved scheduling engine with Dual-MRF readouts and robotics interoperability hooks.

2. Sequence Architecture

2.1 Elliptic Modulo Resonator

Resonator state is defined by the elliptic modular manifold

$$\mathcal{E}_{(q)}: \frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \bmod q,$$

with $(a,b,q)=(1.0,0.64,17)$. Phase evolves by

$$\theta_{n+1}=(3\theta_n+2)\bmod 2\pi,$$

and the RF phase-resolved wave packet is

$$\psi_n(t)=A_n\exp\left(i\left[\omega_n t+\theta_n+\frac{\pi n}{4}\right]\right), \quad n=0,\dots,7.$$

2.2 Dual-MRF Streams

Two synchronous streams are played:

1. Stream A: variable TR/TE/FA schedule for T1-T2 tissue fingerprinting.
2. Stream B: diffusion-perfusion or flow-sensitive schedule for transport biomarkers.

Signal model:

$$s_k = \rho\,f(T_1,T_2,D,f_{flow},\Delta\omega;\,u_k) + \epsilon_k,$$

where u_k contains pulse controls and ϵ_k denotes complex noise.

3. Protocol Blocks

3.1 Cardiac Imaging Block

Variable-cine schedule: TR in 4.8-7.3 ms, TE in 1.9-2.6 ms, ECG and respiratory synchronization.

3.2 Stroke Imaging Block

Diffusion-perfusion fingerprinting with $b\in\{0,150,500,1000,1500\}\,s/mm^2$ and multi-PLD ASL.

3.3 Tumour Resection Robotics Block

Interoperative data exchange uses ROS2 DDS + OpenIGTLink + DICOM-RT metadata, with:

1. Pose update target: 50 Hz.

2. End-to-end latency target: <20 ms.

3. Safety channels: SAR guard, magnet-zone lock, thermal freeze gate.

4. Finite Mathematics Beyond Partial Fourier

Classical partial Fourier recovers missing half-space under conjugate symmetry. Here we define a finite closure operator with fractional boundary gaps.

Let Ω_m be sampled k-space indices and Ω_u unsampled indices. Reconstruction seeks

$$\hat{x}=\arg\min_x \|A_{(\Omega_m)}x-y\|_2^2 + \lambda_1\|Dx\|_1 + \lambda_2\|G_\alpha x\|_2^2.$$

G_α is the fractional boundary-gap operator:

$$(G_\alpha x)_k = x_k - \sum_{j=1}^{p} w_j^{(\alpha)} x_{k-j}, \qquad w_j^{(\alpha)} = (-1)^j \binom{\alpha}{j},$$

with $\alpha=1.35$ and finite stencil order $p=4$.

Boundary-gap decay law:

$$\Delta_g(k)=\frac{g_0}{(k+\varphi)^\alpha}, \qquad g_0=0.12,\ \varphi=1.61803398875.$$

4.1 Worked Numeric Example

For $k=12$,

$$\Delta_g(12)=\frac{0.12}{(12+1.61803398875)^{1.35}} \approx 3.52 \times 10^{-3}.$$

This value is used as the boundary mismatch tolerance for k-space extrapolation at that index.

4.2 Finite-Resolution Gain Estimate

If baseline partial Fourier residual is `r_(PF)=0.018` and proposed residual is `r_(FBG)=0.012`,

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Relative gain=\fracr_(PF)-r_(FBG)r_(PF)=(0.006)/(0.018)=0.333\,(33.3\%).
```

5. Quantum Phase-Resolved Thermometry Coupling

Phase change from temperature follows PRF approximation

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Delta T = \fracDelta\phigamma B_0 alpha_(PRF) TE,
```

with multi-echo robust estimate

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Delta\phi = arg min_(\phi) sum_(e=1)^(E) w_e\,\|s_e-\hats_e(\phi)\|^2,
```

where weights `w_e` are derived from resonator-state confidence and echo SNR.

6. Implementation Artifacts

Generated artifacts:

1. Sequence file: scanner-style pulse sequence with three protocol blocks and reconstruction notes.

2. This technical report with explicit finite derivations and parameter values.

7. Translational Notes

Before deployment on patient workflows:

1. Validate SAR, gradient duty cycle, and PNS envelopes per vendor limits.

2. Verify robotic watchdog and emergency stop with hardware-in-the-loop tests.

3. Benchmark Dual-MRF dictionary mismatch under motion and off-resonance stress tests.

8. Conclusion

The proposed sequence stack unifies cardiac, stroke, and interoperative tumour-resection MRI under one quantum phase-resolved Dual-MRF architecture. Finite fractional boundary-gap operators provide a mathematically explicit extension beyond partial Fourier and can reduce closure residuals while preserving stability in high-acceleration acquisitions.